

Supermarket Sales Dashboard Analysis

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**Domain: Data Analytics & Business Intelligence (Excel + SQL +
Power BI/ Tableau)**

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Task-1: Data Cleaning, Exploratory Data Analysis (EDA) & Basic Business Reporting.

- **Project Overview:**

This project focuses on analyzing supermarket sales data using Excel, SQL, and Power BI. The goal was to identify sales trends, customer behavior, and category performance through data analysis and dashboard visualization.

- **Tools Used:**

Tools	Purpose
Excel	Data Cleaning, Analysis & Pivot Tables
DB Browser (SQLite)	Data Extraction, SQL Queries
Power BI	Dashboard Visualization

- **Excel Analysis:**

- **Summary Metrics:**

Total Sales		2261536.783
Total Orders		9801
Average Order Value		230.7690595

- **Pivot Tables:**

- 1. **Sales By Region**

1. Sales By Region	
Row Labels	Sum of Sales
Central	492646.913
East	669518.726
South	389151.459
West	710219.685
Grand Total	2261536.78

- 2. **Sales By Month:**

2. Sales By Month	
Row Labels	Sum of Sales
2015	479856.208
2016	459436.005
2017	600192.55
2018	722052.019
Grand Total	2261536.78

3. Top 10 Customers:

3. Top 10 Customers	
Row Labels	Sum of Sales
Sean Miller	25043.05
Tamara Chand	19052.218
Raymond Buch	15117.339
Tom Ashbrook	14595.62
Adrian Barton	14473.571
Ken Lonsdale	14175.229
Sanjit Chand	14142.334
Hunter Lopez	12873.298
Sanjit Engle	12209.438
Christopher Conant	12129.072
Grand Total	153811.17

4. Top 5 Products:

4. Top 5 Products	
Row Labels	Sum of Sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.48
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
Grand Total	153385.743

- **SQL Queries & Results:**

1. **Sales By Categories:**

Query:

```
SELECT Category, SUM(Sales) AS total_sales
FROM sales_data
GROUP BY Category;
```

Output:

	Category	total_sales
1	Furniture	728658.5757
2	Office Supplies	705422.334
3	Technology	827455.873

Insight: "Furniture" generated highest sales among all categories.

2. **Sales By Region:**

Query:

```
SELECT Region, SUM(Sales) AS total_sales
FROM sales_data
GROUP BY Region;
```

Output:

	Region	total_sales
1	Central	492646.9132
2	East	669518.726
3	South	389151.459
4	West	710219.6845

Insight: "West" region generated the highest sales among all regions.

3. **Top 5 Customers:**

Query:

```
SELECT "Customer Name", SUM(Sales) AS total_sales
FROM sales_data
GROUP BY "Customer Name"
ORDER BY total_sales DESC
LIMIT 5;
```

Output:

	Customer Name	total_sales
1	Sean Miller	25043.05
2	Tamara Chand	19052.218
3	Raymond Buch	15117.339
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.571

Insight: “Sean Miller” contributed a major portion of total revenue among all customers.

4. Top 5 Products:

Query:

```
SELECT "Product Name", SUM(Sales) AS total_sales
FROM sales_data
GROUP BY "Product Name"
ORDER BY total_sales DESC
LIMIT 5;
```

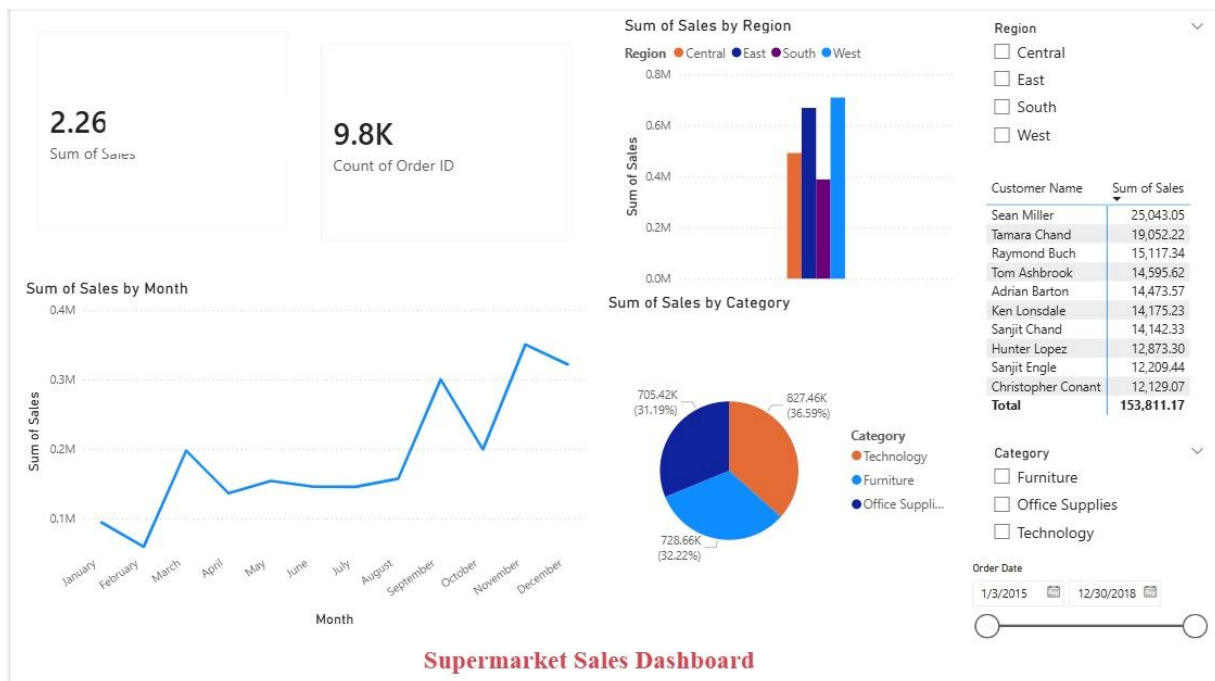
Output:

	Product Name	total_sales
1	Canon imageCLASS 2200 Advanced Copier	61599.824
2	Fellowes PB500 Electric Punch Plasti...	27453.384
3	Cisco TelePresence System EX90 ...	22638.48
4	HON 5400 Series Task Chairs for Big ...	21870.576
5	GBC DocuBind TL300 Electric Binding ...	19823.479

Insight: “Canon imageCLASS 2200 Advanced Copier” contributed a major portion of total revenue among all products.

• Power BI Dashboards:

Screenshot:



[Fig. 1: Sales Dashboard]



[Fig. 2: Filtered Sales Dashboard]

- **Business Insight:**

- **Overall Sales Performance:**

The supermarket achieved strong overall sales performance with total sales of around 2.26M and nearly 9.8K orders. This shows steady customer activity and healthy business growth.

- **Regional Performance:**

The West region generated the highest sales among all regions, followed closely by the East region. This indicates that these regions have stronger customer demand and better sales opportunities.

- **Category Analysis:**

Among all categories, Technology contributed the highest sales revenue. Furniture and Office Supplies also performed well, but Technology remained the top-performing category.

- **Monthly Sales Trend:**

Sales performance changed throughout the year. The business saw noticeable sales growth during September and November, showing higher customer purchases during these months.

- **Customer Insights:**

A small group of top customers contributed a significant portion of total sales. Customers like **Sean Miller** and **Tamara Chand** were among the highest contributors.

- **Filtered Dashboard Observation:**

When filters were applied for:

- South Region
- Technology Category

The dashboard showed focused sales performance for that specific segment. This helps identify how individual regions and categories perform separately.

The filtered dashboard showed that the Technology category performed strongly even within the South region.

Although the total sales value became smaller compared to the complete dashboard, the selected segment still maintained stable sales activity.

The monthly sales trend revealed that customer purchases were not constant throughout the year.

Some months showed higher sales growth, while other months experienced lower sales performance.

This helps identify the periods where demand was stronger.

The filtered customer table also highlighted the top customers contributing to Technology product sales in the South region. This can help businesses identify valuable customers and improve customer-focused strategies.

Using filters makes the dashboard more interactive and useful because it allows businesses to:

analyze specific regions separately,

compare category performance,

track customer behavior,

and identify important sales patterns more clearly.